SKILL-7: REST API - CRUD Operations using ResponseEntity

ID: 2400030128

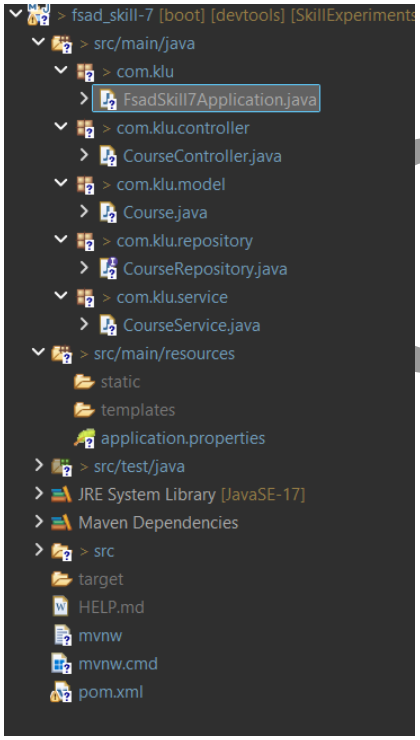
Name: M. Naga Subrahmanyam

Section:2

Scenario:

A university application needs backend services to manage course details such as courseId, title, duration, and fee. The admin should be able to add, update, delete, and view course information. The system must return proper HTTP status codes and structured responses for reliable communication between client and server. Your task is to implement REST endpoints using ResponseEntity to perform complete CRUD operations on course data.

Structure:



0030128

Codes:

Course.java:

```
package com.klu.model;



import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.Table;

@Entity
@Table(name="course_skill7")
public class Course {

  @Id
  private Integer courseId;

  private String title;
  private String duration;
  private Double fee;

  public Course() {}

  public Integer getCourseId() {
    return courseId;
  }

  public void setCourseId(Integer courseId) {
    this.courseId = courseId;
  }

  public String getTitle() {
```

2400030128

```
        return title;
    }

    public void setTitle(String title) {
        this.title = title;
    }

    public String getDuration() {
        return duration;
    }

    public void setDuration(String duration) {
        this.duration = duration;
    }

    public Double getFee() {
        return fee;
    }

    public void setFee(Double fee) {
        this.fee = fee;
    }
}
```

2400030128

CourseController.java:

```
package com.klu.controller;


import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
```

```
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;

import com.klu.model.Course;
import com.klu.service.CourseService;

@RestController
@RequestMapping("/courses")
public class CourseController {

  @Autowired
  CourseService service;

  // Add Course
  @PostMapping
  public ResponseEntity<?> addCourse(@RequestBody Course course) {

    Course saved = service.addCourse(course);

    return new ResponseEntity<>(saved, HttpStatus.CREATED);
  }

  // Get All Courses
  @GetMapping
  public ResponseEntity<List<Course>> getAllCourses() {

    return new ResponseEntity<>(service.getAllCourses(), HttpStatus.OK);
  }

  // Get Course By ID
```

240003012 8

```
@GetMapping("/{id}")
public ResponseEntity<?> getCourse(@PathVariable Integer id) {

    Course course = service.getCourseById(id);

    if(course == null)
        return new ResponseEntity<>("Course Not Found", HttpStatus.NOT_FOUND);

    return new ResponseEntity<>(course, HttpStatus.OK);
}

// Update Course
@PutMapping("/{id}")
public ResponseEntity<?> updateCourse(@PathVariable Integer id, @RequestBody Course
course) {

    Course updated = service.updateCourse(id, course);

    if(updated == null)
        return new ResponseEntity<>("Course Not Found", HttpStatus.NOT_FOUND);

    return new ResponseEntity<>(updated, HttpStatus.OK);
}

// Delete Course
@DeleteMapping("/{id}")
public ResponseEntity<?> deleteCourse(@PathVariable Integer id) {

    boolean deleted = service.deleteCourse(id);

    if(!deleted)
```

2400030128

```
        return new ResponseEntity<>("Course Not Found", HttpStatus.NOT_FOUND);

    return new ResponseEntity<>("Course Deleted Successfully", HttpStatus.OK);
  }

  // Search by Title
  @GetMapping("/search/{title}")
  public ResponseEntity<List<Course>> searchCourse(@PathVariable String title) {

    List<Course> courses = service.searchByTitle(title);

    return new ResponseEntity<>(courses, HttpStatus.OK);
  }
}
```

2400030128

CourseRepository.java:

```
package com.klu.repository;


import java.util.List;

import org.springframework.data.jpa.repository.JpaRepository;

import com.klu.model.Course;

public interface CourseRepository extends JpaRepository<Course, Integer>
{
  List<Course> findByTitle(String title);
}
```

CourseService.java:

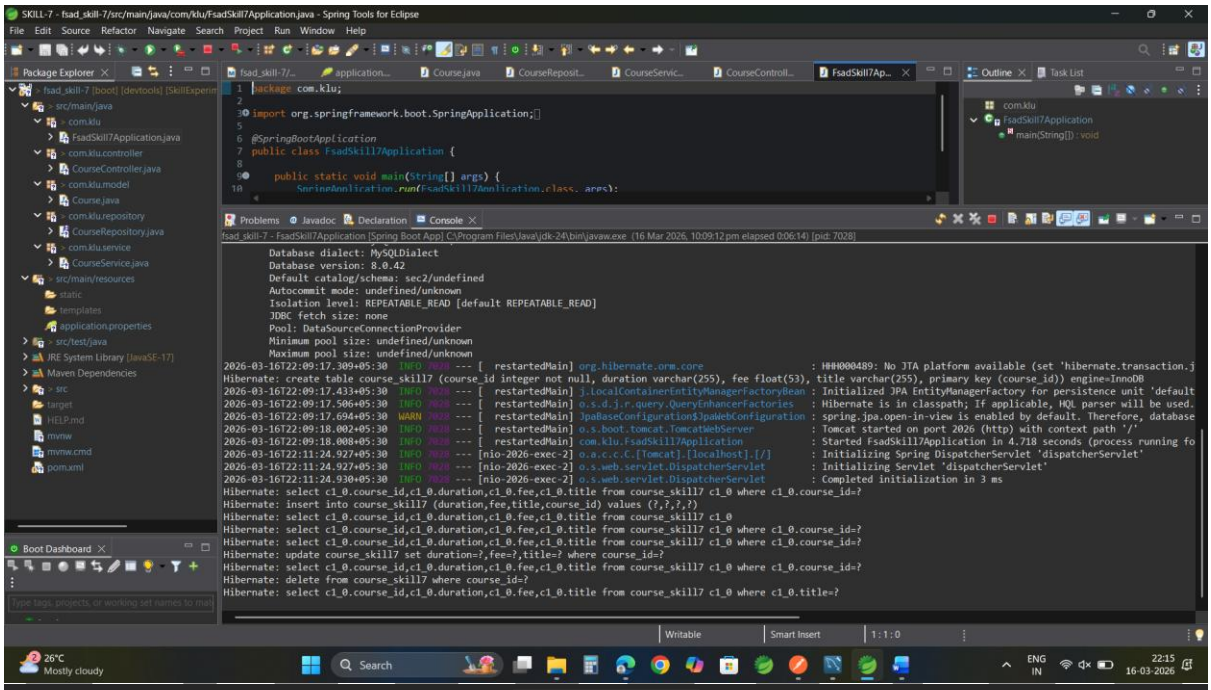
```
package com.klu.service;



import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import com.klu.model.Course;
import com.klu.repository.CourseRepository;

@Service
public class CourseService {

  @Autowired
  CourseRepository repo;

  public Course addCourse(Course course) {
    return repo.save(course);
  }

  public List<Course> getAllCourses() {
    return repo.findAll();
  }

  public Course getCourseById(Integer id) {
    return repo.findById(id).orElse(null);
  }

  public Course updateCourse(Integer id, Course course) {
```

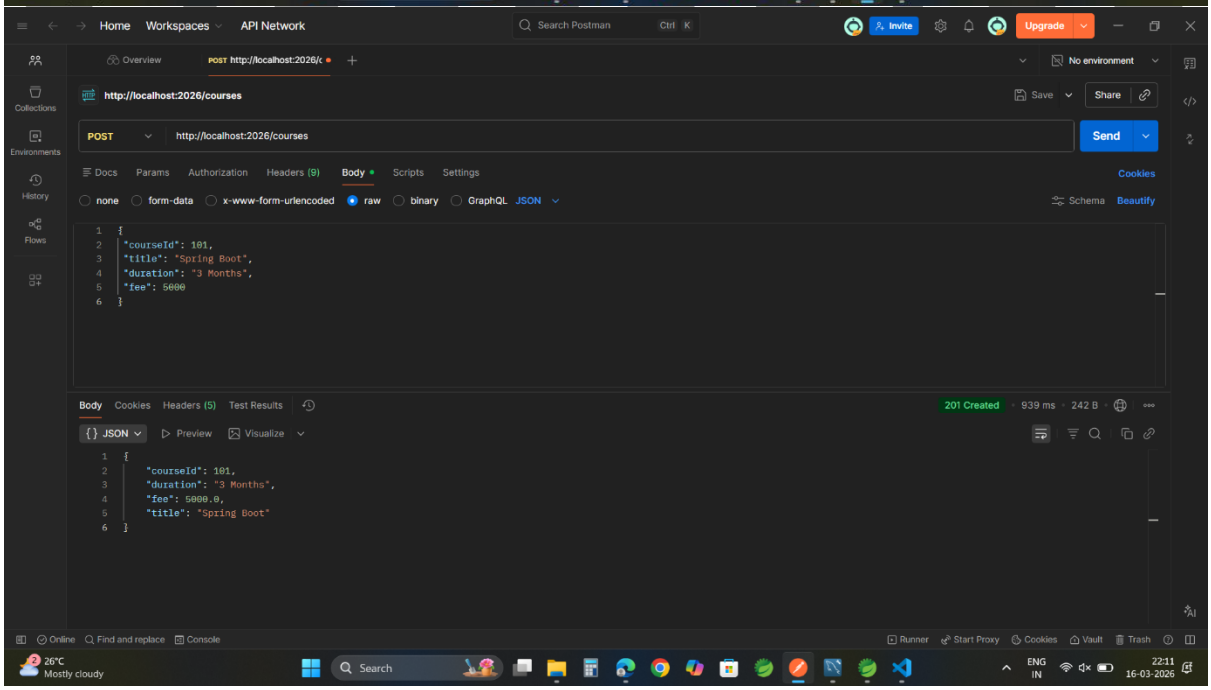
2400030128

```
        Course existing = repo.findById(id).orElse(null);

        if(existing != null) {
            existing.setTitle(course.getTitle());
            existing.setDuration(course.getDuration());
            existing.setFee(course.getFee());
            return repo.save(existing);
        }

        return null;
    }

    public boolean deleteCourse(Integer id) {

        Course course = repo.findById(id).orElse(null);

        if(course != null) {
            repo.delete(course);
            return true;
        }

        return false;
    }

    public List<Course> searchByTitle(String title) {
        return repo.findByTitle(title);
    }
}
```

2400030128

OUTPUTS:


SKILL-7 - fsad_skill-7/src/main/java/com/klu/FsadSkill7Application.java - Spring Tools for Eclipse
package com.klu;
import org.springframework.boot.SpringApplication;
@SpringBootApplication
public class FsadSkill7Application {
public static void main(String[] args) {
Hibernate: create table course_skill7 (course_id integer not null, duration varchar(255), fee float(53), title varchar(255), primary key (course_id)) engine=InnoDB
Hibernate: select c1_0.course_id,c1_0.duration,c1_0.fee,c1_0.title from course_skill7 c1_0 where c1_0.course_id=?
Hibernate: insert into course_skill7 (duration,fee,title,course_id) values (?,?,?,?)
Hibernate: select c1_0.course_id,c1_0.duration,c1_0.fee,c1_0.title from course_skill7 c1_0
Hibernate: update course_skill7 set duration=?,fee=?,title=? where course_id=?
Hibernate: delete from course_skill7 where course_id=?


http://localhost:2026/courses
POST
http://localhost:2026/courses
Send
Body
raw
JSON
{
"courseId": 101,
"title": "Spring Boot",
"duration": "3 Months",
"fee": 5000
}
201 Created
939 ms
242 B
{
"courseId": 101,
"duration": "3 Months",
"fee": 5000.0,
"title": "Spring Boot"
}

Home Workspaces API Network

GET http://localhost:2026/courses

200 OK · 321 ms · 416 B


```
[
    {
        "courseId": 1,
        "duration": "6",
        "fee": 15000.0,
        "title": "FSAD"
    },
    {
        "courseId": 2,
        "duration": "1",
        "fee": 1000.0,
        "title": "HTML"
    },
    {
        "courseId": 3,
        "duration": "2",
        "fee": 5000.0,
        "title": "PYTHON"
    },
    {
        "courseId": 101,
        "duration": "3 Months",
        "fee": 5000.0,
        "title": "Spring Boot"
    }
]
```


Home Workspaces API Network

GET http://localhost:2026/courses/2

200 OK · 32 ms · 221 B


```
{
"courseId": 101,
"title": "Spring Boot",
"duration": "3 Months",
"fee": 5000
}
```

```
{
    "courseId": 2,
    "duration": "1",
    "fee": 1000.0,
    "title": "HTML"
}
```

Home Workspaces API Network

Search Postman Ctrl K

Invite Upgrade

Overview PUT http://localhost:2026/cc

No environment

Collections Environments History Flows

http://localhost:2026/courses/101

Save Share

PUT http://localhost:2026/courses/101 Send

Docs Params Authorization Headers (9) Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "courseId": 101,
3   "title": "Spring Boot",
4   "duration": "4 Months",
5   "fee": 5000
6 }
```

Body Cookies Headers (5) Test Results 200 OK 91 ms 237 B

{} JSON Preview Visualize

```
1 {
2     "courseId": 101,
3     "duration": "4 Months",
4     "fee": 5000.0,
5     "title": "Spring Boot"
6 }
```

Online Find and replace Console Runner Start Proxy Cookies Vault Trash

26°C Mostly cloudy Search ENG IN 22:12 16-03-2026

Home Workspaces API Network

Search Postman Ctrl K

Invite Upgrade

Overview DEL http://localhost:2026/cc

No environment

Collections Environments History Flows

http://localhost:2026/courses/101

Save Share

DELETE http://localhost:2026/courses/101 Send

Docs Params Authorization Headers (9) Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "courseId": 101,
3   "title": "Spring Boot",
4   "duration": "4 Months",
5   "fee": 5000
6 }
```

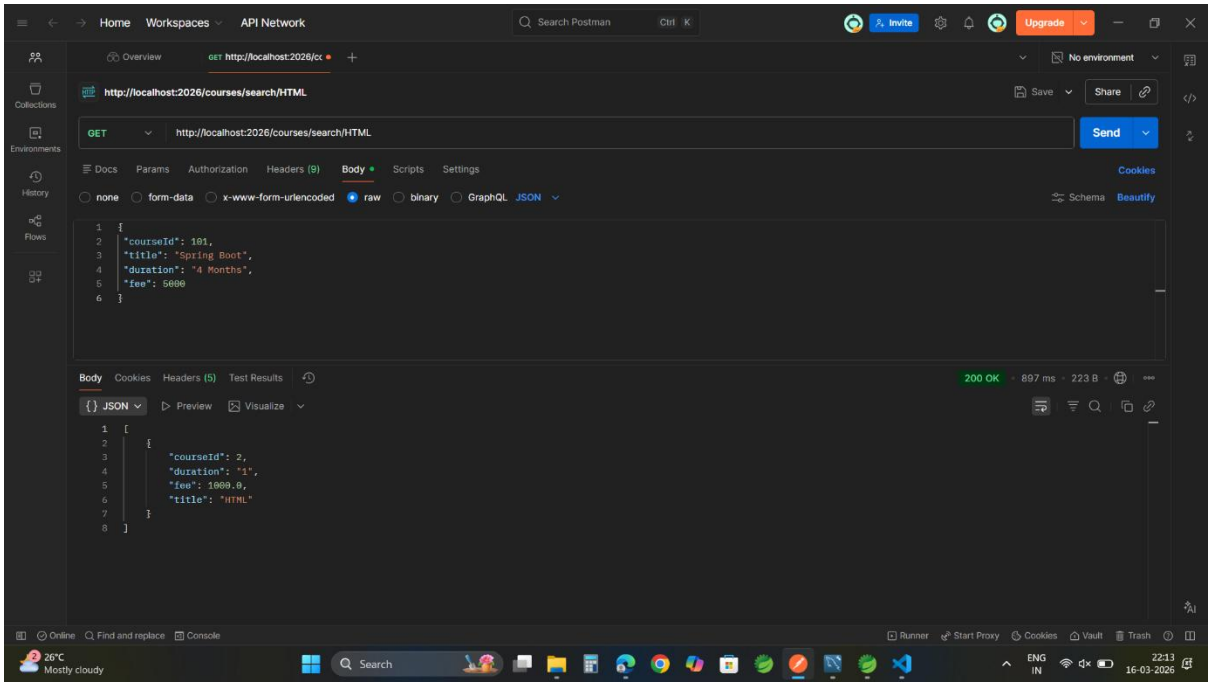
Body Cookies Headers (5) Test Results 200 OK 63 ms 191 B

Raw Preview Visualize

```
1 Course Deleted Successfully
```

Online Find and replace Console Runner Start Proxy Cookies Vault Trash

26°C Mostly cloudy Search ENG IN 22:13 16-03-2026



2400030128

GITHUB LINK:

https://github.com/nagasubrahmanyammeka/SkillExperiments_2400030128_FSAD_Sec2/tree/main/SKILL-7